QUESTIONS FOR CHAPTER 5—PROJECT BUDGETING
问题：第五章——项目预算

1. Usually each of the three principal parties (the **owner**, **designer**, and **contractor**) prepares a **cost estimate** at different times during the life of a project. Describe the **purpose** of each estimate and the **impact** that estimating errors may have on the other two parties.

   通常，三个主要参与方（**业主**、**设计师**和**承包商**）会在项目过程中不同的阶段准备**成本估算**。描述每个估算的**目的**以及估算误差对其他两方可能产生的**影响**。

(1) Owner's Estimate:
- Typically developed **before design** to establish the **maximum funds** to be expended.
- Owner’s estimate includes the **cost of design** and **construction**, plus some **contingency**.
- If the estimate is **too low**, the designer may be asked to produce a **non-achievable project**.
- If the estimate is **too low**, adverse **relations** can develop between all parties.

业主的估算：
- 通常在**设计之前**制定，以确定**最大支出资金**。
- 业主的估算包括**设计和施工的成本**，加上一些**应急费用**。
- 如果估算**过低**，可能会要求设计师完成一个**无法实现的项目**。
- 如果估算**过低**，可能会在各方之间产生不良**关系**。

简化版：

Owner's Estimate: • Developed **before design** • Includes **design, construction costs**, and **contingency** • **Low estimate**: may lead to **unachievable project** or **adverse relationships**
业主的估算 (Owner's Estimate): • **设计前**制定 • 包含**设计、施工成本**和**应急费用** • **估算过低**：可能导致**不可实现的项目**，或**不良关系**

(2) Designer's Estimate:

- The **cost of design** must include work that is not always clearly known in advance.
- Part of the designer's role is to **estimate construction costs** for each **design alternative**.
- **Errors** in the designer's estimate of construction costs may cause disputes with **change orders**.
- Errors in the designer's estimate of construction costs can adversely affect the **owner's economic justification** of the project.

设计师的估算：

- **设计成本**必须包括一些不总是提前明确的工作。
- 设计师的部分职责是为每个**设计备选方案**估算**施工成本**。
- 设计师对施工成本的**估算错误**可能会导致关于**变更单**的争议。
- 设计师对施工成本的估算错误可能会不利于**业主对项目的经济可行性**评估。

简化版：

<table>
<tr><td colspan="2">Designer's Estimate:
Includes unknown work costs
Estimating construction costs is a key responsibility
Errors: may lead to change order disputes and impact economic feasibility</td></tr>
<tr><td>设计师的估算（Designer's Estimate):
包含一些未知工作的成本
估算施工成本为职责之一</td><td>估算错误：可能导致变更单争议，影响经济可行性</td></tr>
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(3) Contractor's Estimate:

- Typically developed **after design** when a full set of **contract documents** has been prepared.
- A reasonable amount of **contingency** should be included to account for **unforeseen work**.
- **Errors** in the contractor's estimate can adversely affect the **contractor's profit**.
- Errors in the contractor's estimate may cause an increase in the number of **change orders**.

承包商的估算：

- 通常在**设计之后**制定，当一套完整的**合同文件**已经准备好时。
- 应包括合理的**应急费用**以应对**不可预见的工作**。
- 承包商估算中的**错误**可能会对**承包商的利润**产生不利影响。
- 承包商估算中的错误可能会导致**变更单数量**的增加。

简化版：

<table>
<tr><td>Contractor's Estimate:
Developed after design is completed
Includes contingency for unforeseen work
Errors: may impact profit and increase change orders</td></tr>
<tr><td>承包商的估算（Contractor's Estimate):
设计完成后制定
包含合理的应急费用以应对意外工作
估算错误：可能影响利润，增加变更单数量</td></tr>
</table>

2. Why is it important to define the **range of accuracy**, in **percentage**, that should be applied to any estimate? Who should **set this range**, and who should be **informed** of the selected range?

为什么定义成本估算的**准确范围**（以**百分比**表示）很重要？谁应当**设定此范围**，并应告知谁**选定的范围**？

Solution:

- A **percent range of accuracy** should be placed on each estimate, so the **user** of the estimate has some concept of the **reliability** that might be reasonably expected of the estimate.
- An **estimate** is only a **forecast** of the cost; therefore, if only a **single number** is provided, people tend to interpret the single number as being **exact** rather than an **approximation**.
- The person **preparing the estimate** should set the **percentage range of accuracy**.
- All **persons using the estimate** should be informed of the **selected range of accuracy**.

- 每个**估算**都应附带一个**准确性的百分比范围**，这样**估算的使用者**可以对估算的**可靠性**有一定的预期。
- **估算**只是对成本的**预测**，因此，如果只提供一个**单一数值**，人们往往会将该数值解读为**精确值**而非**近似值**。
- 由**估算的编制人**来设定**准确性的百分比范围**。
- 所有**使用估算的人**都应被告知**选定的准确性范围**。

简化版：

Reasons for Defining Accuracy Range: • **Percentage range**: helps understand **estimate reliability** • **Estimate as forecast**: single number may be seen as **exact** • **Set by**: **preparer** of the estimate • **Informed**: all **users of the estimate**
定义准确性范围的原因： • **百分比范围**：帮助理解**估算的可靠性** • **估算是预测**：单一数值容易被解读为**精确值** • **设定人**：由**编制人**设定准确性范围 • **知情人**：所有**使用估算的人**应被告知

3. Prepare a **cost estimate** for the construction of a small, **high-quality office building** that contains **18,525 ft²** of floor area. Use the **data in Figure 5-1** to prepare the estimate. Assume the **cost of design** for the project is **7% of construction**, and a **site-work cost** of **$180,000**. What range of **percentage of this cost** would you recommend to define the **level of accuracy**?

为一个**小型高质量办公楼**的**建设**准备**成本估算**，该建筑的**地面面积为 18,525 平方英尺**。使用**图 5-1 中的数据**准备估算。假设**设计成本**为施工成本的 **7%**，场地工程费用为$180,000。你会建议使用多少**百分比范围**来定义此成本的**准确度**？

Solution:

(1) Cost estimate

Known:
1) office building
2) high quality
3) 18,525 SF
4) a site-work cost of $180,000
5) the cost of design for the project is 7% of construction

From Figure 6-2:
total direct costs of $101.15

Since a **medium-quality office building** is specified, use the **medium cost per square foot** from **Figure 6-2**.

Base cost = $101.15/SF × 18,525 SF	=	$	1,873,804
Cost of site-work (given)	=	$	180,000
Total Direct Costs	=	$	2,053,804
Total Direct Costs	=	$	2,053,804
Add design cost, 7% × $2,053,804	=	$	143,766
Estimated Total Cost	=	$	2,197,570

(2) Range of this cost

Since there is no design, use a **range of +50%, -30%**. (Reference: **Page 80** of the book.)

FIGURE 5-1
Illustrative Examples of Cost-per-Square-Foot Information Available from Pricing Manuals.

Component	Office buildings			Motels		
	Low $/SF	Median $/SF	High $/SF	Low $/SF	Median $/SF	High $/SF
Foundation	3.95	4.00	4.80	0.90	1.40	1.60
Floors on grade	3.10	3.15	3.90	3.95	5.00	5.40
Superstructure	14.90	16.90	20.25	10.95	13.30	21.70
Roofing	0.20	0.25	0.30	2.40	3.40	3.45
Exterior walls	4.90	9.75	13.00	2.80	4.45	5.55
Partitions	4.35	5.30	7.05	2.60	3.65	5.25
Wall finishes	2.35	3.75	5.00	0.75	2.60	2.75
Floor finishes	2.05	3.90	5.15	2.40	3.55	4.55
Ceiling finishes	1.55	2.80	3.75	2.05	4.60	4.90
Conveying systems	5.55	6.70	8.25	1.15	1.80	2.35
Specialties	0.65	0.80	2.65	1.10	1.35	4.00
Fixed equipment	1.05	2.80	3.75	1.15	1.65	1.95
Heat/vent/air cond.	8.85	9.50	12.20	3.10	5.55	6.25
Plumbing	3.50	3.80	4.85	4.45	5.40	6.15
Electrical	4.60	4.75	6.25	4.20	7.45	8.20
Total $/SF	$61.55	$78.10	$101.15	$43.95	$65.15	$84.05

Component	Secondary schools			Hospitals		
	Low $/SF	Median $/SF	High $/SF	Low $/SF	Median $/SF	High $/SF
Foundation	1.35	1.85	2.70	4.35	4.80	6.65
Floors on grade	3.65	4.40	6.00	0.30	0.40	0.60
Superstructure	10.95	12.30	17.25	17.05	18.55	25.50
Roofing	1.70	2.05	2.45	3.25	3.70	5.20
Exterior walls	3.75	5.55	8.00	16.00	18.55	25.10
Partitions	5.90	6.55	8.50	7.20	11.00	24.70
Wall finishes	3.05	3.40	5.15	6.75	7.95	11.10
Floor finishes	3.10	3.95	5.25	2.60	2.75	4.00
Ceiling finishes	3.20	3.65	4.65	2.15	2.20	3.55
Conveying systems	0.00	0.00	0.00	12.95	13.00	19.55
Specialties	1.70	1.90	2.60	3.10	3.25	4.60
Fixed equipment	2.85	3.35	6.00	5.20	5.25	7.65
Heat/vent/air cond.	9.05	10.45	14.45	21.65	25.50	36.05
Plumbing	5.05	6.00	9.20	9.10	10.65	16.45
Electrical	10.25	12.00	16.50	13.45	17.50	24.40
Total $/SF	$69.55	$77.40	$108.70	$125.10	$145.05	$215.10

Notes: With no design work, it may range from +50% to −30%. After preliminary design work, it may range from +40% to −20%. On completion of detailed design work, it may range from +15% to −10%.

E4th Ch7Q5:

Prepare a cost estimate for the construction of a small, **median quality, office building** that contains **22,425 square feet** of floor area. Use the data in **Figure 6-2** to prepare the estimate. Assume the cost of design for the project is **8% of construction**, and **a site-work cost of $280,000**. What range or **percentage of this cost** would you recommend to define the **level of accuracy**?

为一个**小型中等质量办公楼**的**建设**准备**成本估算**，该建筑的**地面面积为 22,425 平方英尺**。使用**图 6-2 中的数据**准备估算。假设**设计成本**为施工成本的 **8%**，场地工程费用为$280,000。你会建议使用多少**百分比范围**来定义此成本的**准确度**？

Solution:

(1) Cost estimate

Known:
1) office building
2) median quality
3) 22,425 SF
4) a site-work cost of $280,000
5) the cost of design for the project is 8% of construction

From Figure 6-2:
total direct costs of $136.85

Since a **medium-quality office building** is specified, use the **medium cost per square foot** from **Figure 6-2**.

Base cost = $136.85/SF × 22,425 SF	=	$	3,068,861
Cost of site-work (given)	=	$	280,000
Total Direct Costs	=	$	3,348,861
Total Direct Costs	=	$	3,348,861
Add design cost, 8% × $3,348,861	=	$	267,909
Estimated Total Cost	=	$	3,616,770

(2) Range of this cost

Since there is no design, use a **range of +50%, -30%**. (Reference: **Page 183** of the book.)

FIGURE 6-2 Illustrative examples of cost-per-square-foot information available from pricing manuals.

Component	Office Buildings Low $/SF	Office Buildings Median $/SF	Office Buildings High $/SF	Motels Low $/SF	Motels Median $/SF	Motels High $/SF
Foundation	6.90	7.00	8.40	1.60	2.45	2.80
Floors on grade	5.45	5.50	6.80	6.90	8.75	9.45
Superstructure	26.00	29.60	35.45	19.15	23.30	38.00
Roofing	0.35	0.45	0.55	4.20	5.95	6.05
Exterior walls	8.60	17.05	22.75	4.90	7.80	9.70
Partitions	7.60	9.30	12.35	4.55	6.40	9.20
Wall finishes	4.10	6.55	8.75	1.30	4.55	4.80
Floor finishes	3.60	6.85	9.00	4.20	6.20	7.95
Ceiling finishes	2.70	4.90	6.55	3.60	8.05	8.60
Conveying systems	9.70	11.75	14.45	2.00	3.15	4.10
Specialties	1.15	1.40	4.65	1.95	2.35	7.00
Fixed equipment	1.85	4.90	6.55	2.00	2.90	3.40
Heat/vent/air cond.	15.50	16.65	21.35	5.45	9.70	10.95
Plumbing	6.15	6.65	8.50	7.80	9.45	10.75
Electrical	8.05	8.30	10.95	7.35	13.05	14.35
Total $/SF	107.70	136.85	177.05	76.95	114.05	147.10

Component	Secondary Schools Low $/SF	Secondary Schools Median $/SF	Secondary Schools High $/SF	Hospitals Low $/SF	Hospitals Median $/SF	Hospitals High $/SF
Foundation	2.35	3.25	4.75	7.60	8.40	11.65
Floors on grade	6.40	7.70	10.50	0.55	0.70	1.05
Superstructure	19.15	21.55	30.20	29.85	32.45	44.65
Roofing	3.00	3.60	4.30	5.70	6.50	9.10
Exterior walls	6.55	9.70	14.00	28.00	32.45	43.95
Partitions	10.35	11.45	14.90	12.60	19.25	43.25
Wall finishes	5.35	5.95	9.00	11.80	13.90	19.45
Floor finishes	5.45	6.90	9.20	4.55	4.80	7.00
Ceiling finishes	5.60	6.40	8.15	3.75	3.85	6.20
Conveying systems	0.00	0.00	0.00	22.65	22.75	34.20
Specialties	3.00	3.35	4.55	5.45	5.70	8.05
Fixed equipment	5.00	5.85	10.50	9.10	9.20	13.40
Heat/vent/air cond.	15.85	18.30	25.30	37.90	44.65	63.10
Plumbing	8.85	10.50	16.10	15.95	28.65	28.80
Electrical	17.95	21.00	28.90	23.55	30.65	42.70
Total $/SF	114.85	135.50	190.35	219.00	263.90	376.55

Notes: With no design work, it may range from +50% to −30%. After preliminary design work, it may range from +40% to −20%. On completion of detailed design work, it may range from +15% to −10%.

4. Use the **time and location indices** in **Example 5-2** to estimate the **cost** of a **high-quality office building** that contains **64,500 ft²** of floor area. The building is to be **constructed 2 years** from now in **city A**. The cost of a **similar building** that contains **95,000 ft²** was completed last year in **city C** for a cost of **$7,790,000**.

   使用**示例 5-2 中的时间和位置指数**来估算一个**高质量办公楼**的**成本**，该建筑的**地面面积为 64,500 平方英尺**。该建筑将在 **2 年后**在 **A 市**建造。去年，**C 市**建成了一个**地面面积为 95,000 平方英尺**的**类似建筑**，其成本为$7,790,000。

Solution:

Known:

1) Previous cost: $7,790,000, completed last year in City C.
2) n = To be constructed 2 years from now + completed last year = 3 years.
3) time adjustment = $(1+i)^3 = (1.0183)^3$
4) Location indices: City A = 1,025 City C = 1,260
5) Size: City A = 64,500 SF City C = 95,000 SF

$$\begin{aligned}
\text{Estimated Cost} &= \text{Previous Cost} \times \text{Adjustments (time} \times \text{location} \times \text{size)} \\
&= \$7{,}790{,}000 \times (1.0183)^3 \times (1{,}025/1{,}260) \times (64{,}500/\ 95{,}000) \\
&= \$7{,}790{,}000 \times (1.0559) \times (0.8135) \times (0.6789) \\
&= \$4{,}542{,}799
\end{aligned}$$

Example 5-2

Use the time and location indices below to calculate the forecast cost for a building with 62,700 SF of floor area. The building is to be constructed 3 years from now in city B. A similar type building that cost $2,945,250 and contained 38,500 SF was completed 2 years ago in city D.

Year	Index
3 yr ago	358
2 yr ago	359
1 yr ago	367
Current year	378

Location	Index
City A	1025
City B	1170
City C	1260
City D	1240

An equivalent compound interest can be calculated based on the change in the cost index during the 3-year period:

$$\frac{378}{358} = (1+i)^3 \rightarrow i = 1.83\%$$

Estimated cost	Previous cost	Time adjustment	Location adjustment	Size adjustment
"	= $2,945,250 ×	(1 + 0.0183)⁵ ×	(1170/1240) ×	(62,700/38,500)
"	= $2,945,250 ×	1.095 ×	0.944 ×	1.629
"	= $4,959,403			

E4th Ch6Q8:

Use the time and location indices in **Example 6-4** to estimate the cost of a building that contains **73,500 square feet** of floor area. The building is to be constructed three years from now in **City B**. The cost of a similar type building that contains **105,000 square feet** was completed last year in **City D** for a cost of **$15,175,000**.

使用**示例 5-2 中的时间和位置指数**来估算一栋建筑的**成本**，该建筑的**地面面积为 73,500 平方英尺**。该建筑将在 **3 年后**在 **B 市**建造。去年，**D 市**建成了一个**地面面积为 105,000 平方英尺的类似建筑**，其成本为$15,175,000。

Solution:

Known:

1) Previous cost: $15,175,000, completed last year in City D.
2) n = To be constructed 3 years from now + completed last year = 4 years.
3) time adjustment = $(1+i)^4 = (1.071)^4$
4) Location indices: City B = 1,805 City D = 2,091
5) Size: City B = 73,500 SF City D = 105,000 SF

$$
\begin{aligned}
\text{Estimated Cost} &= \text{Previous Cost} \times \text{Adjustments (time} \times \text{location} \times \text{size)} \\
&= \$15{,}175{,}000 \times (1.071)^4 \times (1805/2091) \times (73{,}500/105{,}000) \\
&= \$15{,}175{,}000 \times (1.3157) \times (0.8632) \times (0.7000) \\
&= \$12{,}064{,}103
\end{aligned}
$$

Example 6-4 Use the time and location indices below to calculate the forecast cost for a building with 78,400 square feet of floor area. The building is to be constructed 3 years from now in city C. A similar type building that cost $3,828,800 and contained 40,400 square feet was completed 4 years ago in city D.

Year	Index
4 years ago	4,374
3 years ago	5,028
2 years ago	5,199
1 year ago	5,461
Current year	5,761

Location	Index
City A	1,646
City B	1,805
City C	1,911
City D	2,091
City E	1,876

There is no method to predict future cost escalation with absolute certainty. For this example, a compound interest is calculated based on the trend in cost indices of the past 4 years. Using the compound interest equation $F/P = (1 + i)^n$, an interest rate can be calculated as follows:

$$\frac{5761}{4374} = (1+i)^4 \rightarrow i = 7.1\%$$

The combined adjustments of costs for time, location, and size can be calculated as follows:

Estimated cost		Previous cost		Time adjustment		Location adjustment		Size adjustment
"	=	$3,828,800	×	$(1.071)^7$	×	(1911/2091)	×	(78,400/40,400)
"	=	$3,838,800	×	1.6163	×	0.9139	×	1.9406
$11,004,000								

Example 6-4 Information

Example 6-4 provides **the time** and **location indices** needed to adjust costs based on time and geographical location. Specifically:

1) **Time Indices:** These are provided for different years to calculate a **time adjustment** rate based on compound interest.
2) **Location Indices:** Different location indices allow for cost adjustments based on the city where the building is constructed.
3) **Adjustment Formula:** The example uses these indices with a compound interest formula to calculate adjustments for time, location, and size.

5. During the **feasibility study** of a project, the initial **estimated cost** for **design and construction** is **$3.7 million**. It is anticipated that the **cost to maintain and operate** the facility after completion of construction will be **$250,000 per year**. Assuming the owner must obtain a **return on the initial investment of 15%**, what **net annual income** must be received to **economically justify** the project? Assume no **salvage value** after using the facility for **12 years**.

在项目的**可行性研究**中，**设计和施工的初始估计成本**为$3.7 百万。预计在施工完成后，设施的**维护和运营成本**将为每年$250,000。假设业主必须获得**初始投资回报率为 15%**，那么项目必须产生多少**年净收入**才能在**经济上合理**？假设使用设施 **12 年后**没有**残值**。

Solution:

According to the Capital Recovery Formula (资本回收公式):

Capital Recovery Amount (资本回收年金), $A_{CR} = P \times \frac{i(1+i)^n}{(1+i)^n-1}$

** 在这个公式中，**A** 表示**每年的等额回收款项**（也可以理解为年金支付额），用于将初始投资现值 **P** 转换为一系列等额的年支付。这通常用于计算在一定利率 i 和年限 n 下，需要的年支付额 **A** 以回收初始投资 **P**。这时的 **A** 仅指年金支付的金额。

According to the Annual Income Formula (年收入公式):

Annual Income (年收入) = $A_{CR} + A = P \cdot \frac{i(1+i)^n}{(1+i)^n-1} + A$

where A = Annual Expense 或 Fixed Annual Cost

** 在这个公式中，**A** 表示**每年的额外成本或收入**，而非年金支付额。这里的 **A** 通常是指**维护和运营成本**或其他每年固定支出的金额。

Known:

1) the initial estimated cost (初始投资), P = $3.7 million = $3,700,000
2) the initial investment (年利率), i = 15% = 0.15
3) the useful life (使用年限), n = 12 years
4) the cost to maintain and operate the facility per year (每年的维护和运营成本), A = $250,000

Determine the A_{CR}/P ratio (15%, 12 years):

$$A_{CR}/P = \frac{i(1+i)^n}{(1+i)^n-1} = \frac{0.15\,(1+0.15)^{12}}{(1+0.15)^{12}-1} = 0.1844808$$

Substitute the formula to calculate annual income:

$$\begin{aligned}\text{Annual Income} &= A_{CR} + A \\ &= P \cdot \frac{i(1+i)^n}{(1+i)^n-1} + A \\ &= \$3{,}700{,}000 \times 0.1844808 + \$250{,}000 \\ &= \$682{,}579 + \$250{,}000 \\ &= \$932{,}579 \text{ per year}\end{aligned}$$

The project would need net revenues of $932,579 per year to be economically justified.

E4th Ch7Q2:

During the **feasibility study** of a project, the initial **estimated cost** for **design and construction** is **$6.5 million**. It is anticipated that the cost to maintain and operate the facility after completion of construction will be **$530,000 per year**. Assuming the owner must obtain a **return on the initial investment of 14%**, what **net annual income** must be received to **economically justify** the project. Assume no **salvage value** after using the facility for **15 years**.

在项目的**可行性研究**中，**设计和施工的初始估计成本**为$6.5 百万。预计在施工完成后，设施的**维护和运营成本**将为每年**$530,000**。假设业主必须获得**初始投资回报率为 14%**，那么项目必须产生多少**年净收入**才能在**经济上合理**？假设使用设施 **15 年后**没有**残值**。

Solution:

According to the Capital Recovery Formula (资本回收公式) :

Capital Recovery Amount (资本回收年金) , $A_{CR} = P \times \frac{i(1+i)^n}{(1+i)^n-1}$

** 在这个公式中，**A** 表示**每年的等额回收款项**（也可以理解为年金支付额），用于将初始投资现值 **P** 转换为一系列等额的年支付。这通常用于计算在一定利率 i 和年限 n 下，需要的年支付额 **A** 以回收初始投资 **P**。这时的 **A** 仅指年金支付的金额。

According to the Annual Income Formula (年收入公式) :

Annual Income (年收入) = $A_{CR} + A = P \cdot \frac{i(1+i)^n}{(1+i)^n-1} + A$

where A = Annual Expense 或 Fixed Annual Cost

** 在这个公式中，**A** 表示**每年的额外成本或收入**，而非年金支付额。这里的 **A** 通常是指**维护和运营成本**或其他每年固定支出的金额。

Known:

1) the initial estimated cost (初始投资), P = $6.5 million = $6,500,000
2) the initial investment (年利率), i = 14% = 0.14
3) the useful life (使用年限), n = 15 years
4) the cost to maintain and operate the facility per year (每年的维护和运营成本) , A = $530,000

Determine the A_{CR}/P ratio (14%, 15 years):

$$A_{CR}/P = \frac{i(1+i)^n}{(1+i)^n-1} = \frac{0.14\,(1+0.14)^{15}}{(1+0.14)^{15}-1} = 0.1628090$$

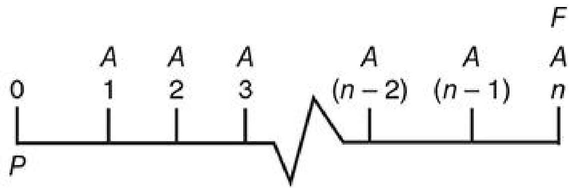
Substitute the formula to calculate annual income:

$$\begin{aligned}
\text{Annual Income} &= A_{CR} + A \\
&= P \cdot \frac{i(1+i)^n}{(1+i)^n-1} + A \\
&= \$6{,}500{,}000 \times 0.1628090 + \$530{,}000 \\
&= \$1{,}059{,}000 + \$530{,}000 \\
&= \$1{,}590{,}000 \text{ per year}
\end{aligned}$$

The project would need net revenues of $1,590,000 per year to be economically justified.

Fundamental Equations of the Time Value of Money (Reference in Page 235)

P = Present Worth
F = Future Sum
A = Equal Payment Series
i = Annual Interest Rate
n = Study Period (years)



Single Payment Series				
		Equation	**Factor form**	**Equation number**
Compound Amount	$F =$	$P[(1+i)^n]$	$P(F/P,i,n)$	(7-1)
Present Worth	$P =$	$F\left[\frac{1}{(1+i)^n}\right]$	$F(P/F,i,n)$	(7-2)
Equal Payment Series				
		Equation	**Factor form**	**Equation number**
Compound Amount	$F =$	$A\left[\frac{(1+i)^n-1}{i}\right]$	$A(F/A,i,n)$	(7-3)
Sinking Fund	$A =$	$F\left[\frac{i}{(1+i)^n-1}\right]$	$F(A/F,i,n)$	(7-4)
Present Worth	$P =$	$A\left[\frac{(1+i)^n-1}{i(1+i)^n}\right]$	$A(P/A,i,n)$	(7-5)
Capital Recovery	$A =$	$P\left[\frac{i(1+i)^n}{(1+i)^n-1}\right]$	$P(A/P,i,n)$	(7-6)

The factor form designation represents an abbreviated form of the equation when using factor tables and is illustrated as: (F/P,i,n) means find the Future Sum, F, from a known Present Worth, P, and Interest Rate, i, during a Study Period of n.

6. The **cost estimate** of a project is **$3.5 million**. **Annual costs** for **maintaining and operating** the facility are forecast as **$250,000 per year**. After **8 years**, it is anticipated that the facility will be **sold for $2.0 million**. If the **owner requires a 15% return** on their investment, what **net annual income** must be received to **recover the capital investment** of the project?

   某项目的**成本估算**为$3.5 百万。设施的**年度维护和运营成本**预测为**每年$250,000**。预计在 **8 年后**，该设施将以$2.0 百万**出售。如果业主**要求 15%的投资回报率，该项目需要多少**年净收入**才能**收回资本投资**？

Solution:

According to the Capital Recovery Formula (资本回收公式)：

Capital Recovery Amount (资本回收年金)，$A_{CR} = P \times \frac{i(1+i)^n}{(1+i)^n-1}$

According to the Sinking Fund Formula (终值回收公式)：

Sinking Fund Amount (终值回收金，也可记为 Annual Deposit 年存款)，$A_{SF} = F \times \frac{i}{(1+i)^n-1}$

** 这个系数用于将一个未来值 (F) 转换为每年的等额存入金额 (A)，通常用于计算每年需要的存款，以便在最终年限内达到目标终值。

According to the Annual Income Formula (年收入公式)：

Annual Income (年收入) = $A_{CR} + A + A_{SF} = P \cdot \frac{i(1+i)^n}{(1+i)^n-1} + A - F \cdot \frac{i}{(1+i)^n-1}$

Known:

1) the initial estimated cost (初始投资), P = $3.5 million = $3,500,000
2) the initial investment (年利率), i = 15% = 0.15
3) the useful life (使用年限), n = 8 years
4) the cost to maintain and operate the facility per year (每年的维护和运营成本), A = $250,000
5) estimated selling price after 8 years (8 年后的预计出售价格), F = $2.0 million = $2,000,000

Determine the A_{CR}/P ratio (15%, 8 years):

$$A_{CR}/P = \frac{i(1+i)^n}{(1+i)^n-1} = \frac{0.15\,(1+0.15)^8}{(1+0.15)^8-1} = 0.2228501$$

Determine the A_{SF}/F ratio (15%, 8 years):

$$A_{SF}/F = \frac{i}{(1+i)^n-1} = \frac{0.15}{(1+0.15)^8-1} = 0.0728501$$

Substitute the formula to calculate annual income:

$$\begin{aligned}\text{Annual Income} &= A_{CR} + A + A_{SF}\\ &= P \cdot \frac{i(1+i)^n}{(1+i)^n-1} + A - F \cdot \frac{i}{(1+i)^n-1}\\ &= \$3{,}500{,}000 \times 0.2228501 + \$250{,}000 - \$2{,}000{,}000 \times 0.0728501\\ &= \$779{,}975 + \$250{,}000 - \$157002\\ &\approx \$873{,}000 \text{ per year}\end{aligned}$$

The project would need net revenues of $873,000 per year to be economically justified.

E4th Ch7Q3:

The cost estimate of a project is $4.8 million. Annual costs for maintaining and operating the facility are forecast as $230,000 per year. After 9 years, it is anticipated the facility will be sold for $3.9 million. If the owner requires a 12% return on investment, what net annual income must be received to recover the capital investment of the project?

某项目的**成本估算**为$4.8 百万。设施的**年度维护和运营成本**预测为**每年$230,000**。预计在 **9 年后**，该设施将以$3.9 百万**出售。如果业主**要求 12%的投资回报率，该项目需要多少**年净收入**才能**收回资本投资**？

Solution:

According to the Capital Recovery Formula (资本回收公式)：

Capital Recovery Amount (资本回收年金)， $A_{CR} = P \times \frac{i(1+i)^n}{(1+i)^n-1}$

According to the Sinking Fund Formula (终值回收公式)：

Sinking Fund Amount (终值回收金，也可记为 Annual Deposit 年存款)， $A_{SF} = F \times \frac{i}{(1+i)^n-1}$

** 这个系数用于将一个未来值 (F) 转换为每年的等额存入金额 (A)，通常用于计算每年需要的存款，以便在最终年限内达到目标终值。

According to the Annual Income Formula (年收入公式)：

Annual Income (年收入) = $A_{CR} + A + A_{SF} = P \cdot \frac{i(1+i)^n}{(1+i)^n-1} + A - F \cdot \frac{i}{(1+i)^n-1}$

Known:

1) the initial estimated cost (初始投资), P = $4.8 million = $4,800,000
2) the initial investment (年利率), i = 12% = 0.12
3) the useful life (使用年限), n = 9 years
4) the cost to maintain and operate the facility per year (每年的维护和运营成本)，A = $230,000
5) estimated selling price after 9 years (9 年后的预计出售价格)，F = $3.9 million = $3,900,000

Determine the A_{CR}/P ratio (12%, 9 years):

$$A_{CR}/P = \frac{i(1+i)^n}{(1+i)^n-1} = \frac{0.12\,(1+0.12)^9}{(1+0.12)^9-1} = 0.1876789$$

Determine the A_{SF}/F ratio (12%, 9 years):

$$A_{SF}/F = \frac{i}{(1+i)^n-1} = \frac{0.12}{(1+0.12)^9-1} = 0.0676789$$

Substitute the formula to calculate annual income:

$$\begin{aligned}\text{Annual Income} &= A_{CR} + A + A_{SF} \\ &= P \cdot \frac{i(1+i)^n}{(1+i)^n-1} + A - F \cdot \frac{i}{(1+i)^n-1} \\ &= \$4{,}800{,}000 \times 0.1876789 + \$230{,}000 - \$3{,}900{,}000 \times 0.0676789 \\ &= \$900{,}859 + \$230{,}000 - \$263{,}948 \\ &\approx \$867{,}000 \text{ per year}\end{aligned}$$

The project would need net revenues of $867,000 per year to be economically justified.

7. The **initial investment** for a project is **$4.7 million**. A **net annual profit** of **$1.5 million** is anticipated. Using a **12% desired rate of return** on the investment, what is the **payback period** for the project?

   某项目的**初始投资**为$4.7 百万。预计每年产生**净利润**为$1.5 百万。如果期望的**投资回报率为 12%**，该项目的**回收期**是多少？

Solution:

Use the Present Worth to Annuity Formula (现值-年金公式）to calculate the payback period:

Capital Recovery Amount (资本回收年金) , $A_{CR} = P \times \frac{i(1+i)^n}{(1+i)^n-1}$

→ Present Value (现值，也可以理解为初始投资或现期价值), $P = A_{CR} \times \frac{(1+i)^n-1}{i(1+i)^n}$

** P 通常表示在未来一系列等额年金（A）给定的情况下，等价的现期一次性投资金额，即在当前需要投入的资金量。
A 表示每年的等额支付金额（或净利润）。

Known:
1) the initial estimated cost (初始投资), P = $4.7 million = $4,700,000
2) capital recovery amount (资本回收年金，每年净利润)，A_{CR} = $1,500,000
3) the initial investment (年利率), i = 12% = 0.12
4) the useful life (使用年限，回收期), n = ？ years

Substitute the known values and adjust the formula to find n:

$$P/A_{CR} = \frac{(1+i)^n-1}{i(1+i)^n} = \frac{(1+0.12)^n-1}{0.12(1+0.12)^n} \rightarrow (1+i)^n = \frac{1}{1-\left(\frac{P}{A_{CR}}\right)\cdot i} \rightarrow (1+0.12)^n = \frac{1}{1-\left(\frac{4.7}{1.5}\right)\cdot 0.12}$$

$$(1.12)^n = \frac{1}{1-\left(\frac{4.7}{1.5}\right)\cdot 0.12} = \frac{1}{1-0.376} = \frac{1}{0.624} = 1.6025641$$

Take the logarithm of both sides of the equation to solve for n:

$$log((1.12)^n) = log(\,1.6025641)$$
$$n \cdot log(1.12) = log(\,1.6025641)$$

Compute the logarithm and solve for n:

$$n \cdot 0.0492180 = 0.2048154$$
$$n = \frac{0.2048154}{0.0492180} = 4.16 \text{ yr} = 4 \text{ years} + 0.16 \cdot 12 \text{ months} \approx 4 \text{ years and } 2 \text{ months}$$

The project has a payback period of 4.16 years.
This means that it will take about 4 years and 2 months for the project to achieve the expected 14% ROI and recover the initial investment.

E4th Ch7Q4:

The **initial investment** for a project is **$6.8 million**. A **net annual profit** of **$1.8 million** is anticipated. Using a **14% desired rate of return** on the investment, what is the **payback period** for the project?

某项目的**初始投资**为$6.8 百万。预计每年产生**净利润**为$1.8 百万。如果期望的**投资回报率为 14%**，该项目的**回收期**是多少？

Solution:

Use the Present Worth to Annuity Formula (现值-年金公式) to calculate the payback period:

Capital Recovery Amount (资本回收年金)， $A_{CR} = P \times \frac{i(1+i)^n}{(1+i)^n-1}$

→ Present Value (现值，也可以理解为初始投资或现期价值), $P = A_{CR} \times \frac{(1+i)^n-1}{i(1+i)^n}$

** P 通常表示在未来一系列等额年金（A）给定的情况下，等价的现期一次性投资金额，即在当前需要投入的资金量。
A 表示每年的等额支付金额（或净利润）。

Known:

1) the initial estimated cost (初始投资), P = \$6.8 million = \$6,800,000
2) capital recovery amount (资本回收年金，每年净利润)， A_{CR} = \$1,800,000
3) the initial investment (年利率), i = 14% = 0.14
4) the useful life (使用年限，回收期), n = ？ years

Substitute the known values and adjust the formula to find n:

$$P/A_{CR} = \frac{(1+i)^n-1}{i(1+i)^n} = \frac{(1+0.14)^n-1}{0.14(1+0.14)^n} \rightarrow (1+i)^n = \frac{1}{1-\left(\frac{P}{A_{CR}}\right)\cdot i} \rightarrow (1+0.14)^n = \frac{1}{1-\left(\frac{6.8}{1.8}\right)\cdot 0.14}$$

$$(1.14)^n = \frac{1}{1-\left(\frac{6.8}{1.8}\right)\cdot 0.14} = \frac{1}{1-0.5289} = \frac{1}{0.4711} = 2.1226916$$

Take the logarithm of both sides of the equation to solve for n:

$$log((1.14)^n) = log(2.1226916)$$
$$n \cdot log(1.14) = log(2.1226916)$$

Compute the logarithm and solve for n:

$$n \cdot 0.0569049 = 0.3268869$$

$$n = \frac{0.3268869}{0.0569049} = 5.74 \text{ yr} = 5 \text{ years} + 0.74 \cdot 12 \text{ months} \approx 5 \text{ years and } 9 \text{ months}$$

The project has a payback period of 5.74 years.
This means that it will take about 5 years and 9 months for the project to achieve the expected 14% ROI and recover the initial investment.